

National University of Science & Technology

Institute of Geographical Information Systems

Digital Mapping and Image Processing

Lab#3

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Task 1

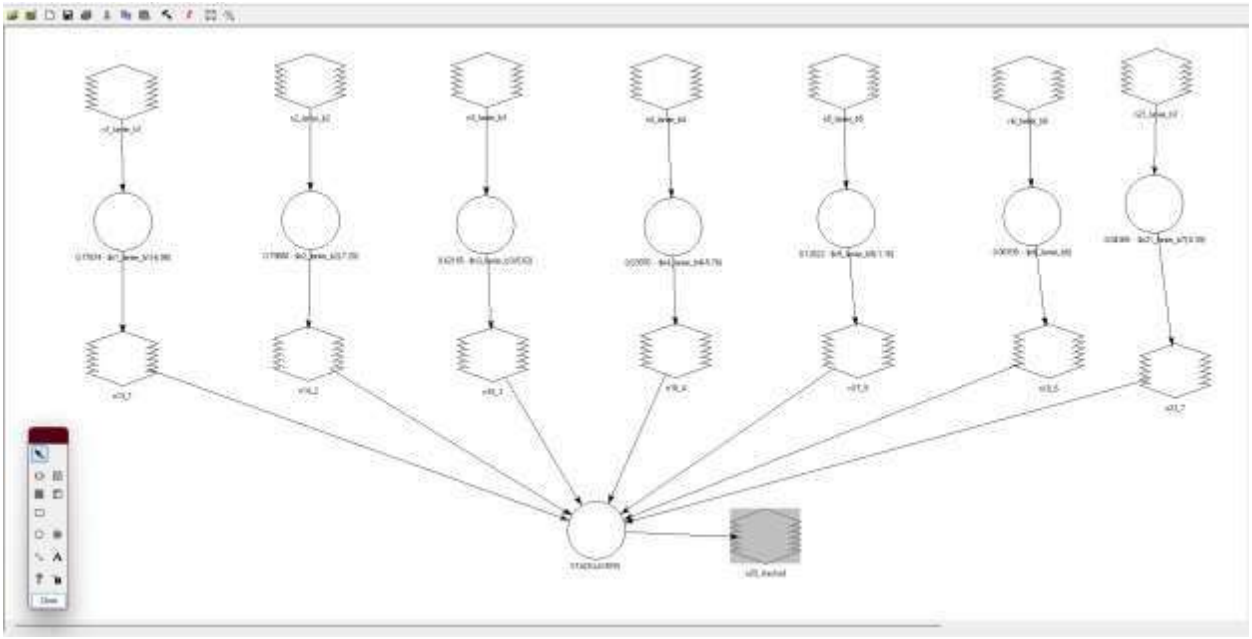


Fig 1.1. Modell maker (all bands DN values changing to radiance and stacking simultaneously)

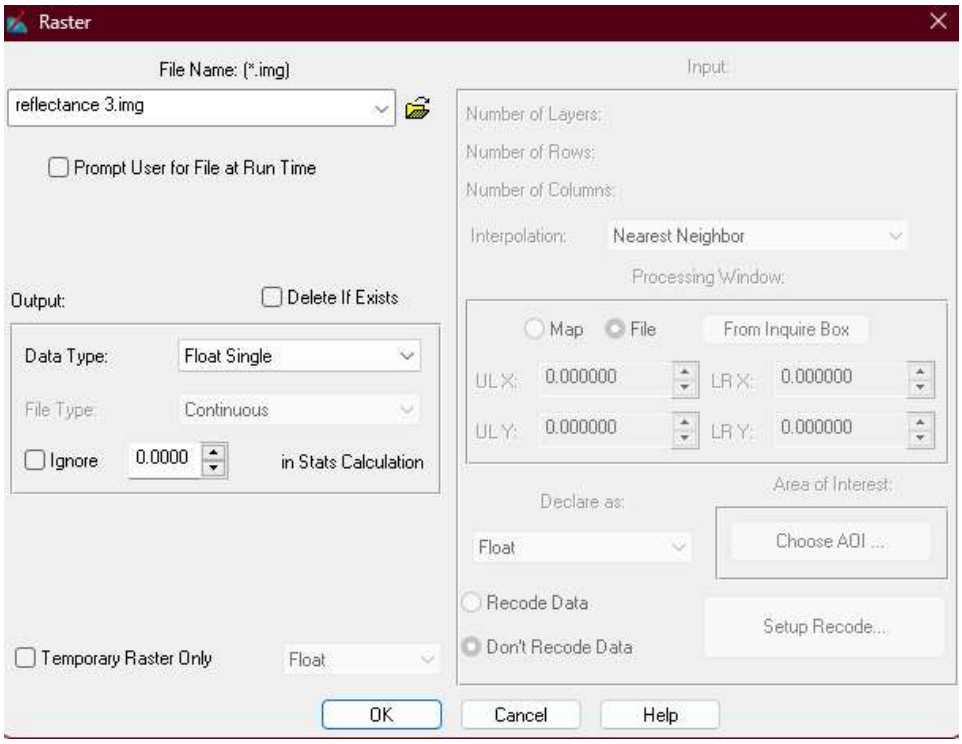


*Fig 1.2. Radiance output at stage 3 of model maker (formula used : [DN * Gain]+Bias)*

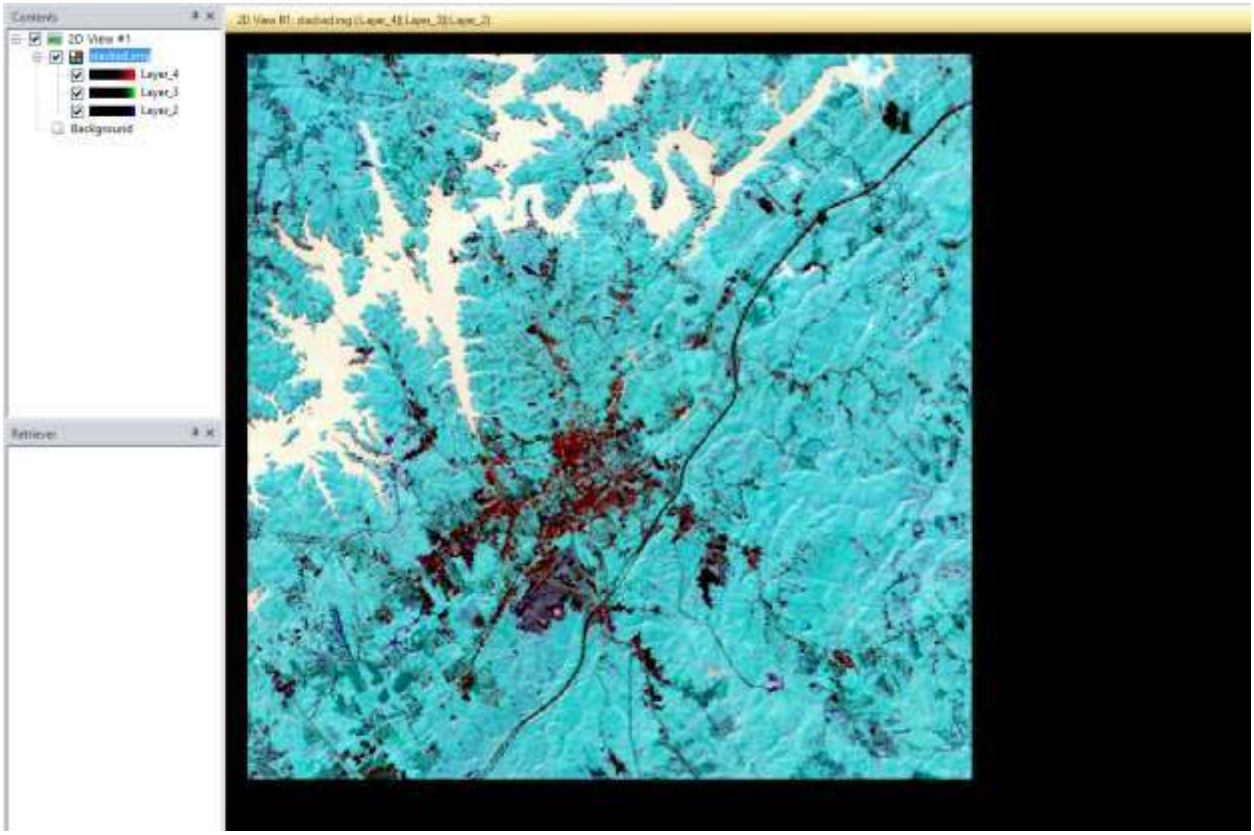


Fig 1.3. Output radiance and Stacked image

Pixel data for stacked.img(:Layer_1)

Row	0	1	2
0	-074.201	-076.201	-080.201
1	-074.201	-078.201	-077.201
2	-075.201	-081.201	-080.201
3	-078.201	-079.201	-080.201
4	-080.201	-082.201	-083.201
5	-082.201	-082.201	-081.201
6	-082.201	-082.201	-078.201
7	-081.201	-081.201	-081.201
8	-075.201	-082.201	-078.201
9	-074.201	-078.201	-079.201
10	-072.201	-072.201	-074.201
11	-070.201	-072.201	-073.201
12	-071.201	-073.201	-075.201

Close Help

Fig 1.4. Pixel data for radiance stacked image

Pixel data for lanier.img(:Layer_1)

Row	0	1	2	3	4	5	6
0	68	70	74	77	77	82	82
1	68	72	71	75	78	77	77
2	69	75	74	74	74	73	73
3	72	73	74	76	79	74	74
4	74	76	77	77	76	72	75
5	76	76	75	75	75	76	77
6	76	76	72	76	77	76	74
7	75	75	75	76	75	76	73
8	69	76	72	82	78	72	73
9	68	72	73	78	75	68	70
10	66	66	68	66	68	70	74
11	64	66	67	65	67	68	72
12	65	67	69	68	68	67	68

Close Help

Fig 1.5. Pixel data for actual stacked image

Task 2

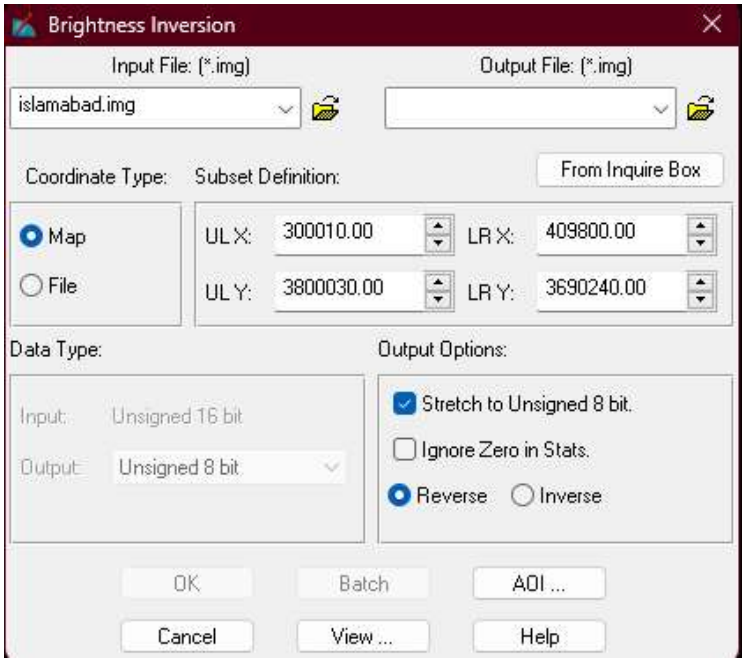


Fig 2.1. Brightness inversion and reverse tab

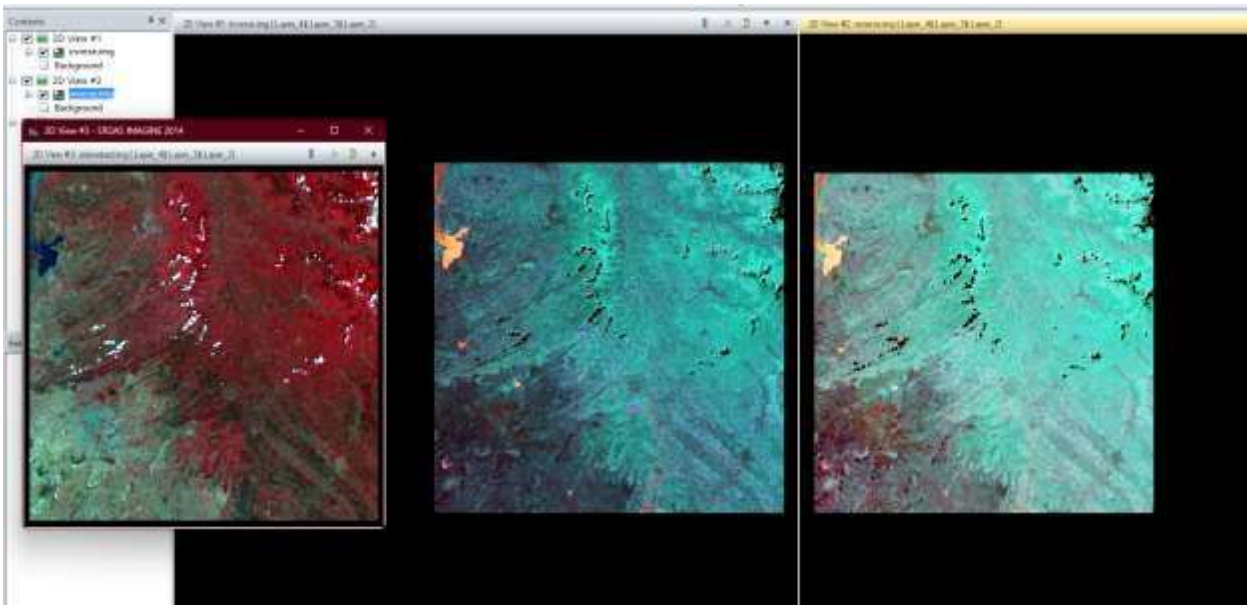
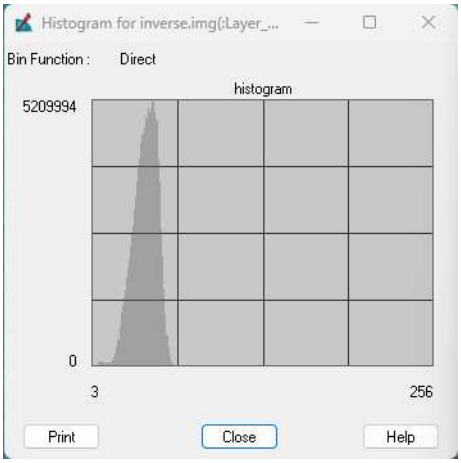
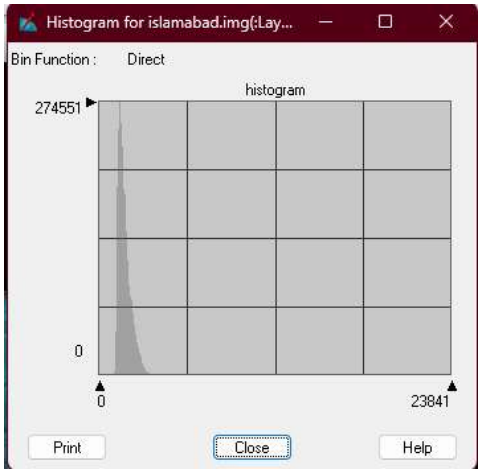
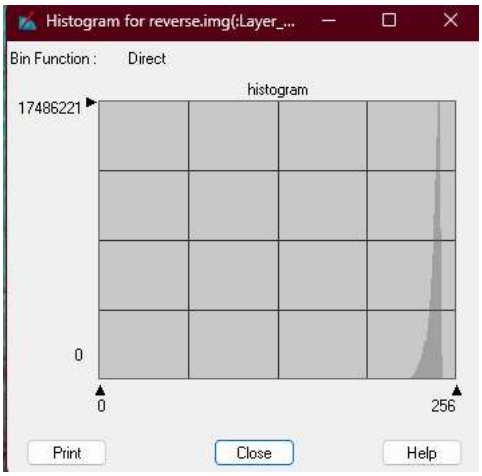


Fig 2.2. actual img, inverse img and reverse image in order left to right







Inverse Image:

- Pixel values are mathematically transformed—dark becomes bright, bright becomes dark
- Creates a true photographic negative of the scene
- Bright red vegetation turns greenish-cyan
- Dark water bodies and shadows become bright white or cyan
- Urban and barren areas shift to pinkish or purplish tones
- Histogram flips completely: values near 0 move to near 255
- Pixel data is permanently altered

Reverse Image:

- Pixel values remain unchanged
- Only the display stretch is flipped
- Produces a faded, washed-out negative appearance
- Red vegetation appears as pale cyan
- Water bodies show as medium tones instead of extremes
- Urban areas display as dull grey-green
- Histogram stays identical to the original
- Only the visual interpretation of brightness is reversed
- Creates a subtler effect compared to inversion

Histogram Equalization

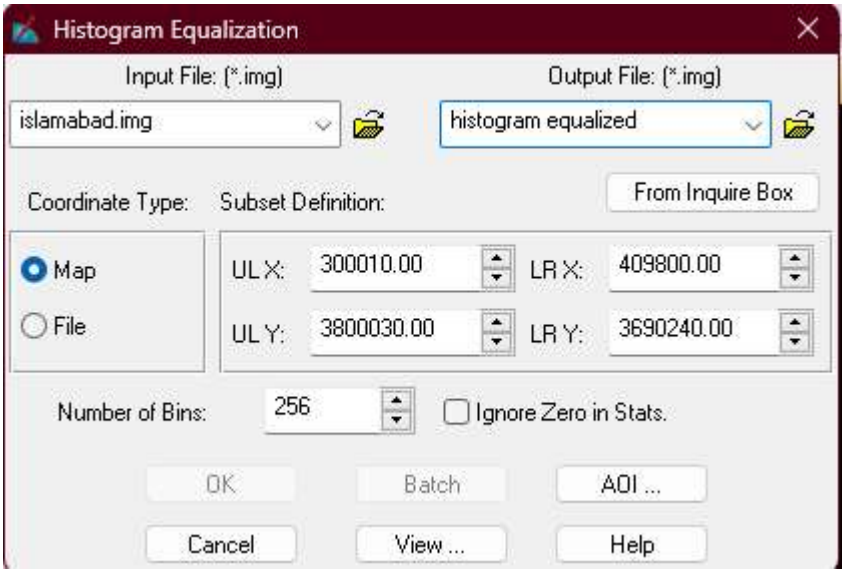


Fig 2.6.1. Histogram equalization tab

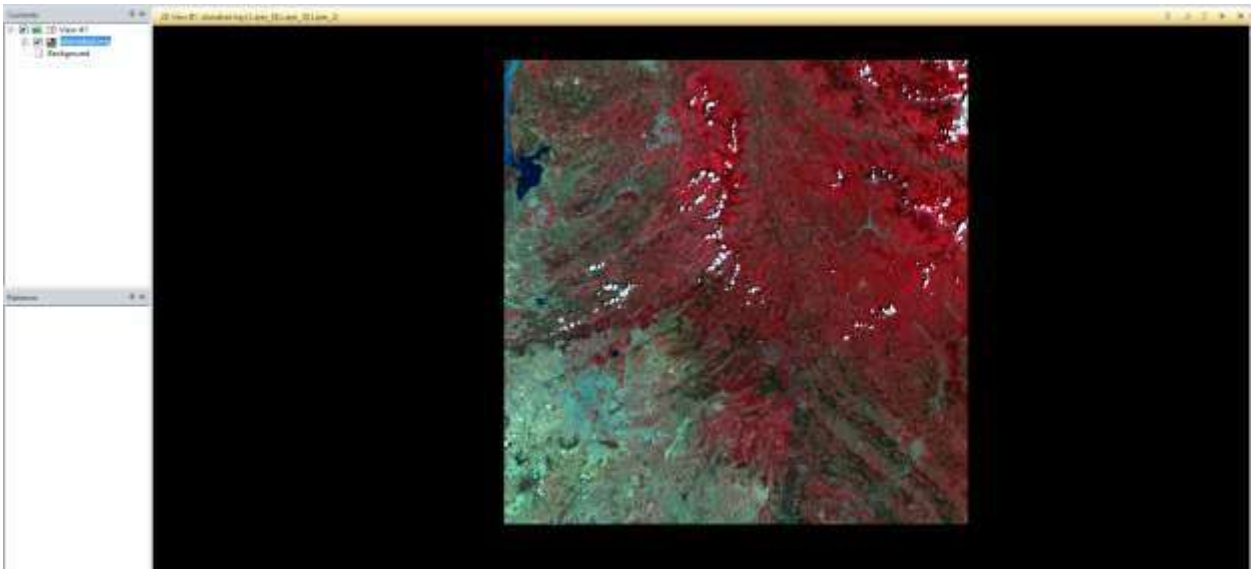


Fig 2.6.2. Islamabad.img

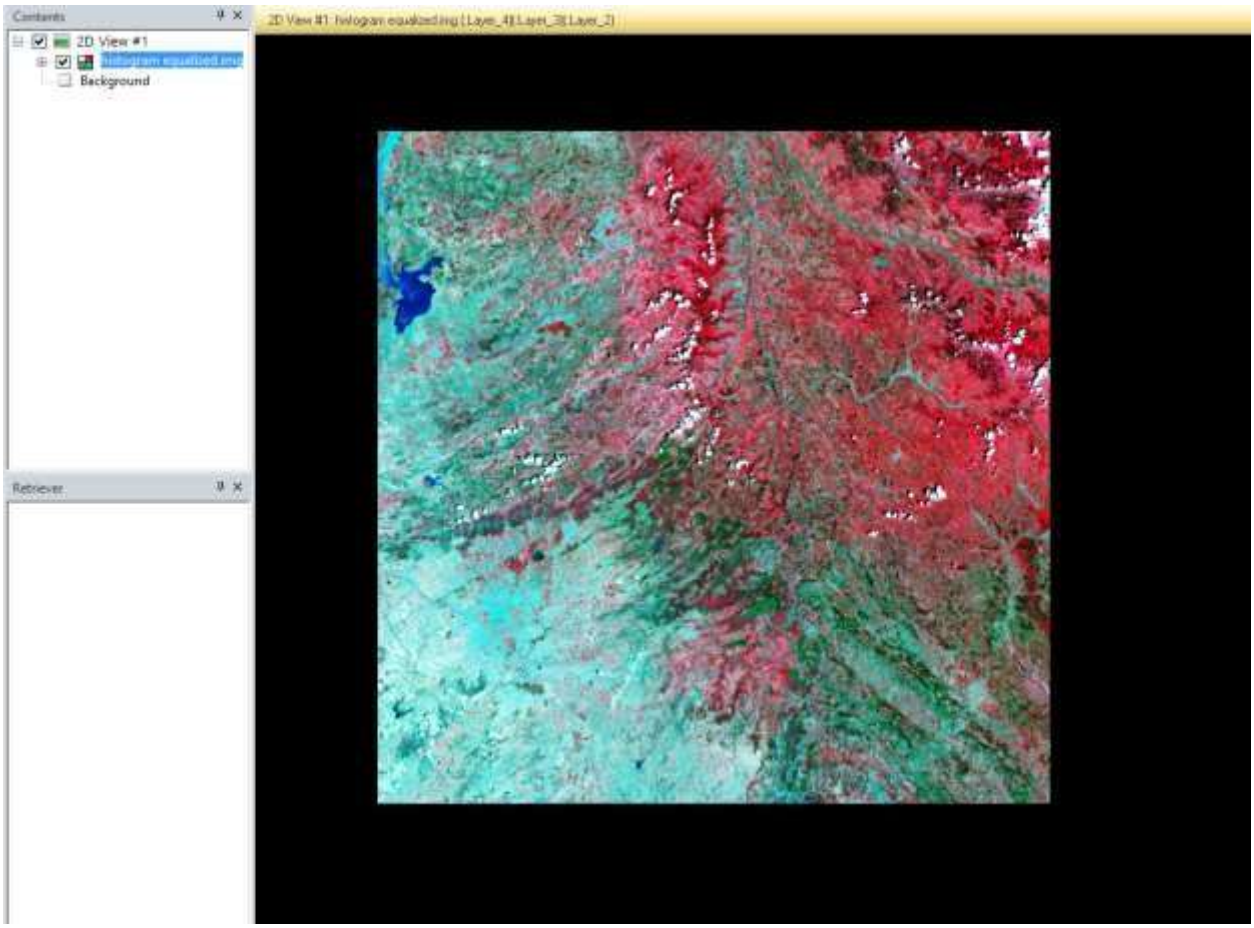


Fig 2.6.3. Histogram equalized image of Islamabad.img